

Optimizing and Adapting Language Models for Domain-Specific Tasks

Advanced RAG & Parameter-Efficient Fine-Tuning

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Enterprise Challenges in LLM Deployment

Two fundamental gaps when deploying Generative AI:

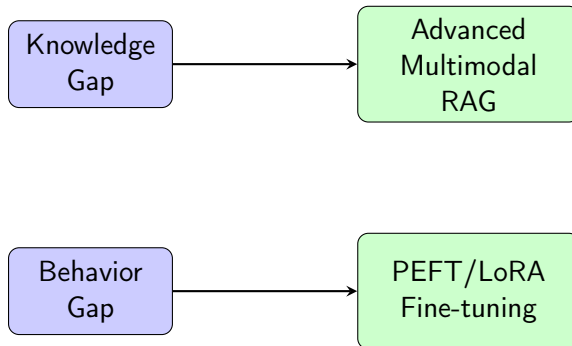
Knowledge Gap

- Static training data
- Hallucination risks
- No access to proprietary documents

Behavior Gap

- Inconsistent communication style
- Cannot follow domain workflows
- Out-of-character drift

How can we bridge these gaps with resource-constrained deployment?



Two Complementary Solutions

- **RAG Pipeline:** Accurate retrieval from complex documents (text, tables, images)
- **Fine-tuned SLMs:** Behavioral consistency at low cost

RAG Background: Basic Pipeline

Indexing Phase (Offline)

- 1 Document Loading
- 2 Chunking
- 3 Embedding
- 4 Vector Storage

Query Phase (Online)

- 1 Query embedding
- 2 Similarity search
- 3 Context augmentation
- 4 LLM generation

Standard approach: Dense retrieval with simple text chunking

Limitations of Simple RAG

Three Critical Issues

- ❶ **Structure Loss:** Tables and layouts destroyed during parsing
 - Financial reports, medical forms become unreadable
- ❷ **Monolingual Bias:** MiniLM optimized for English
 - Poor performance on French documents (GPM use case)
- ❸ **"Lost in the Middle":** LLMs struggle with long contexts
 - Relevant info buried in retrieved chunks

Simple RAG: 0.648 correctness, 0.654 context precision

Three-Pillar Design

① Structure Preservation

- UnstructuredIO: Maintains table structure, extracts images

② Semantic Enrichment

- Multimodal summarization (text + tables + images)
- Store both raw + summarized content

③ Multi-layer Retrieval

- Hybrid Search: BM25 (keyword) + Dense (semantic)
- Cross-Encoder Reranking: Prevent "Lost in the Middle"

Embedding: multilingual-e5-large-instruct (best French performance < 1B params)

GPM Financial Reports

Challenge:

- Complex tables with nested structure
- French language documents
- Multi-page financial statements

Task:

- Extract specific figures
- Answer accounting queries

CCAM Medical Coding

Challenge:

- 9,000+ medical procedure codes
- Precise code retrieval required
- Clinical terminology

Task:

- Find correct CCAM code
- Provide code descriptions

Evaluation Framework: RAGAS Metrics

LLM-as-a-Judge Methodology

Two judge models: gemma-3-12b-it, mistral-small-24b-instruct-2501

Metric	Definition
Answer Correctness	Semantic similarity: generated vs ground-truth
Answer Relevancy	How well response addresses query
Faithfulness	Response grounded in retrieved context
Context Precision	Relevant chunks ranked highly
Domain-Specific	
Code Hit Rate@k	Correct code in top-k results (CCAM)

All metrics: continuous scale $[0, 1]$, averaged across two judges

RAG Evaluation Results

Table: GPM Test Set (25 questions)

Metric	Simple	Advanced
Correctness	0.648	0.976
Relevancy	0.788	0.972
Faithfulness	0.732	0.940
Context Prec.	0.654	0.960
Improvement	—	+32.8%

Table: CCAM Test Set (10 scenarios)

Metric	Simple	Advanced
Hit Rate@5	0.4	0.8
Hit Rate@20	0.6	0.9
Improvement	—	+40%

Key Takeaway

Structure preservation + reranking mechanism → **32.8% correctness gain**

Why Not Just Prompt Engineering?

Limitations of In-Context Learning

- 1 **Attention Dilution:** Long system prompts → degraded performance
- 2 **Inconsistency:** Out-of-character drift in extended conversations
- 3 **Cost:** Large models (GPT-4) expensive for continuous deployment

Theoretical Foundation: Intrinsic Dimensionality

- Behavioral patterns lie in a **low-dimensional subspace**
- Parameter-efficient methods (LoRA) can capture this subspace
- Small models sufficient to **reproduce** expert behavior

"Behavioral expertise is compressible"

Low-Rank Adaptation (LoRA)

Inject trainable low-rank matrices into frozen model:

$$W' = W_{\text{frozen}} + \underbrace{BA}_{\text{trainable}}$$

where $W \in \mathbb{R}^{d \times k}$, $B \in \mathbb{R}^{d \times r}$, $A \in \mathbb{R}^{r \times k}$, and $r \ll \min(d, k)$

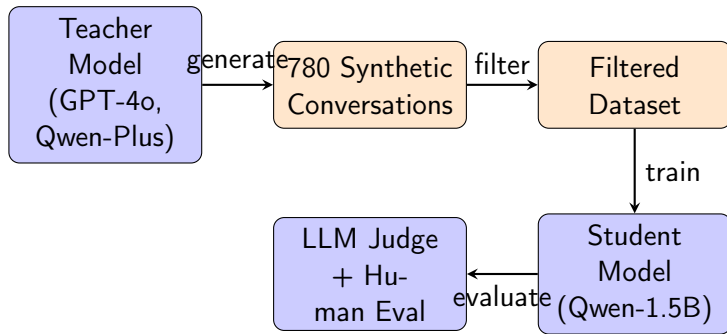
Parameter Count:

- Full FT: Update all $d \times k$ params
- LoRA: Only $r(d + k)$ params
- Typical: $r = 16$, reduction $\sim 99\%$

Our Configuration:

- Rank $r = 16$
- Applied to: Attention + **FFN** layers
- Hypothesis: Behavior requires both "seeing" and "processing"

Knowledge Distillation Pipeline



Three-Stage Process

- 1 **Generation:** Teacher models create expert conversations
- 2 **Training:** Student learns behavioral patterns via supervised fine-tuning
- 3 **Evaluation:** Multi-dimensional assessment (LLM + Human)

Case Study: Tarot Reader Chatbot

Three-Layer Complexity

① Knowledge Layer

- 78 cards (upright + reversed)

② Operational Layer

- Ritual workflow: Greeting → Explore → Draw → Interpret → Synthesize

③ Emotional Layer

- Empathetic language
- Open-ended questions
- Maintain character consistency

[Demo Screenshot]

User draws 3 cards,
receives personalized reading

Why good test bed? Behavioral complexity >> knowledge complexity

Model Selection & Training Configuration

Table: Performance on Standard Benchmarks

Model	MMLU (Knowledge)	GSM8K (Reasoning)	MATH (Hard Logic)
Llama-3.2-1B-Instruct	49.3	44.4	30.6
Gemma-2-2.6B-Base	52.2	30.3	25.3
Qwen2.5-0.5B-Instruct	47.5	49.6	34.4
Qwen2.5-1.5B-Instruct	60.9	73.2	55.2

Why Qwen2.5?

- Best performance/parameter ratio
- Strong multilingual capability
- Apache 2.0 license

LoRA Configuration:

- Rank: 16
- Target: Attention + FFN
- Trainable params: 1.2% of total

Fine-tuning Results

Table: LLM-as-a-Judge Scores

Model	Avg Score	Win Rate
GPT-4 (prompt)	0.85	25%
Qwen-1.5B (FT)	0.88	71%
Qwen-0.5B (FT)	0.79	—
Qwen-1.5B (LoRA)	0.88	—
Qwen-1.5B (Full)	0.88	—

Key Findings:

- LoRA $\approx$ Full FT (loss 1.15 vs 1.12)
- Style Adherence: +6%
- Workflow Compliance: +17%

Table: Operational Metrics

Metric	GPT-4	Qwen-1.5B
Cost (per 1K queries)	\$15	\$1.5
Latency (TTFT)	400ms	<100ms
Deployment	API only	Local
Cost Reduction	—	90%

Breakthrough

Behavioral consistency surpassing GPT-4 at 10% cost

Two Fundamental Principles

① "Behavioral expertise is compressible"

- Large models needed to *generate* expert behavior
- Small models sufficient to *reproduce* it
- Behavioral patterns $\in$ low-dimensional subspace

② RAG + Fine-tuning = Complementary Framework

- RAG: Dynamic knowledge, grounded responses
- Fine-tuning: Behavioral consistency, style adherence
- Together: Knowledge-accurate **AND** behaviorally reliable

Enterprise AI without sacrificing quality

Limitations & Future Directions

Limitations

- LLM judges may have style biases
- Limited human evaluation (30 convs, 2 annotators)
- Synthetic data may miss edge cases
- Single-domain testing (Tarot)

Future Work

- 1 **RLHF:** Refine from real user interactions
- 2 **Multilingual:** Leverage Qwen's native capabilities
- 3 **Hybrid Architecture:** RAG + Fine-tuned SLMs
- 4 **Cross-Domain:** Medical, legal, financial chatbots

Goal: Generalize distillation framework to diverse enterprise applications

Thank You!

Questions?

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